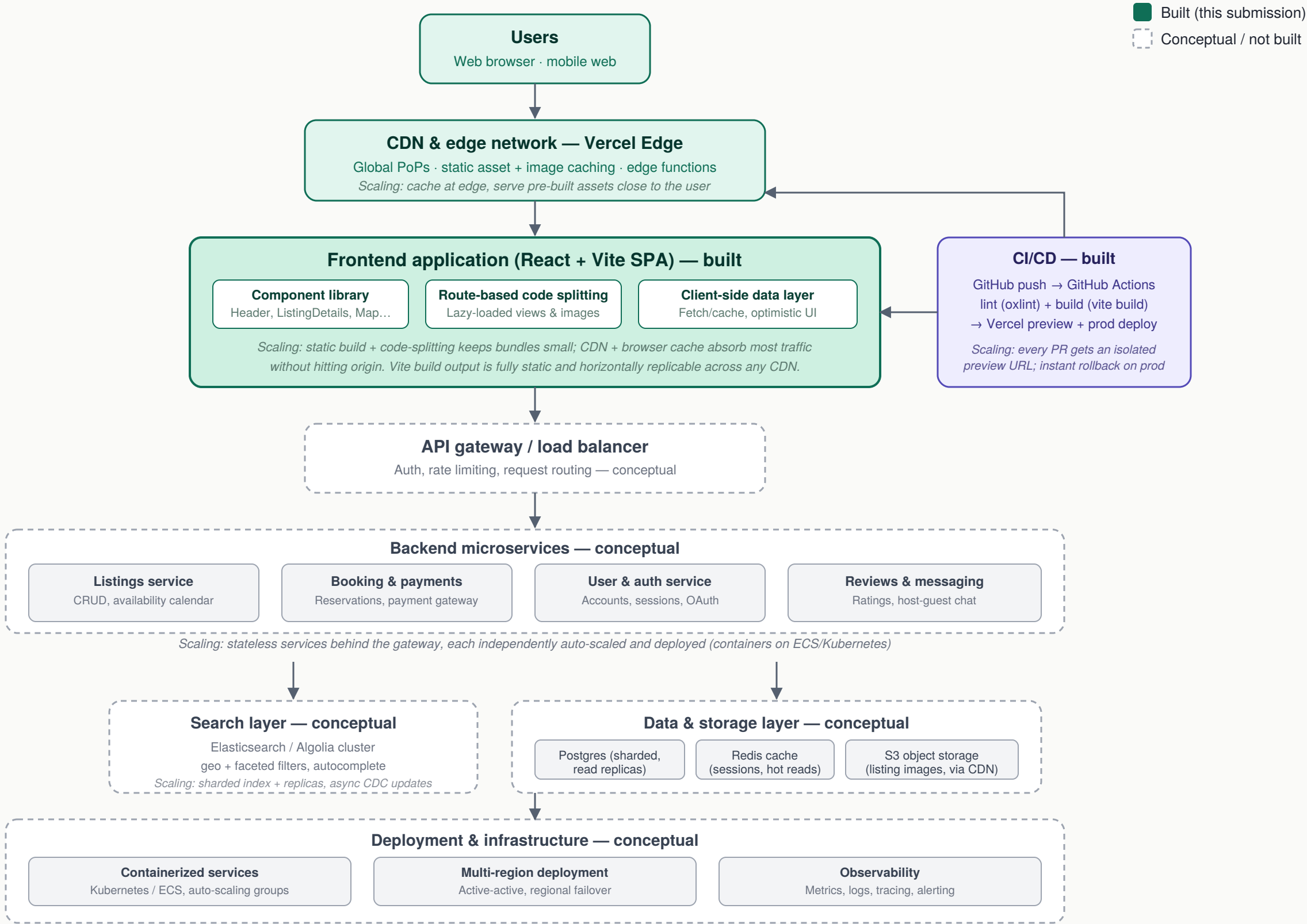


Vacation-Rental Marketplace — High-Level Architecture & Scaling Strategy

Frontend implemented (React + Vite, deployed on Vercel) · Backend/Storage/Search/Infra shown conceptually per assignment scope



Scaling strategy summary

Frontend (built):

Static Vite build served from Vercel's global CDN · route-level code splitting and lazy-loaded images keep first-load payload small
· traffic spikes are absorbed at the edge, not at any origin server · every PR gets its own preview deployment for fast, safe iteration.

Backend (conceptual):

Stateless microservices behind an API gateway, each scaled independently based on load; listings/search reads dominate traffic so those services get more replicas than write-heavy booking/payment services.

Storage & search (conceptual):

Postgres sharded by region with read replicas for the high read:write ratio typical of a listings marketplace; Redis in front for hot reads (session, popular listings); search is offloaded to a dedicated Elasticsearch/Algolia cluster kept in sync via async events, so complex geo/filter queries never hit the primary database.